Class #4 - Solution Set of a system

1. Fill in the blanks: A system of linear equations is consistent if and only if an echelon form of its __________matrix does __________have a row of the form

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & \square \end{bmatrix}.$$

2. Existence and Uniqueness Theorem:
A linear system is consistent if and only if the rightmost column of the *augmented* matrix is __________a pivot column.
If a linear system is consistent, then the solution set contains either
   (a) a unique solution, when there are no __________variables (that is, every column *except* the rightmost column is a __________column, or
   (b) infinitely many solutions, when there is at least one free variable.

3. Example: Find the general solution of the linear system whose augmented matrix has been reduced to

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}.$$

This requires a back-substitution to find a solution.

Its reduced echelon form is:

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}.$$

In this case, the solution can be read off.

4. Def: Let A, B be sets. A rule that assigns exactly __________element in B for __________element in A is called a *function (or Transformation or Mapping)* from A to (or into) B.
   We write
   $$f : A \to B$$
   to mean that f is a function from A into B.
   The set A is called the __________of f, and B is called the __________of f.

5. Def: For an $x \in A$, $f(x)$ is called the __________of x under f.
   The *Range* of f is the set of all __________as x varies over the set A:
   Range$(f) := \{f(x) \in$ __________ $: x \in$ __________$\}$.

6. Recall $\mathbf{R^n}$:_________________________, where n is a __________number.
   This ordered n-tuple of real numbers $(u_1, u_2, \cdots, u_n)$ __________called a *row-vector*.
   The same thing written in columns
   $$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$
   is called a __________vector or just a vector. So,
   $$\mathbf{R^n} = \{(u_1, u_2, \cdots, u_n) : u_i \in \mathbf{R}\} = \left\{ \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} : u_i \in \mathbf{R} \right\}.$$

7. Def: The vector $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ is called the __________vector in $\mathbf{R^n}$.
   Example: Given a nonzero vector in $\mathbf{R^5}$:__________.